Key Findings

- The penetration of semiconductor components classified as IoT is expected to grow from 7% in 2019 to 12% by 2025.
- Four key components are driving this IoT semiconductor growth: Microcontrollers, connectivity chipsets, AI chipsets, and security chipsets and modules.

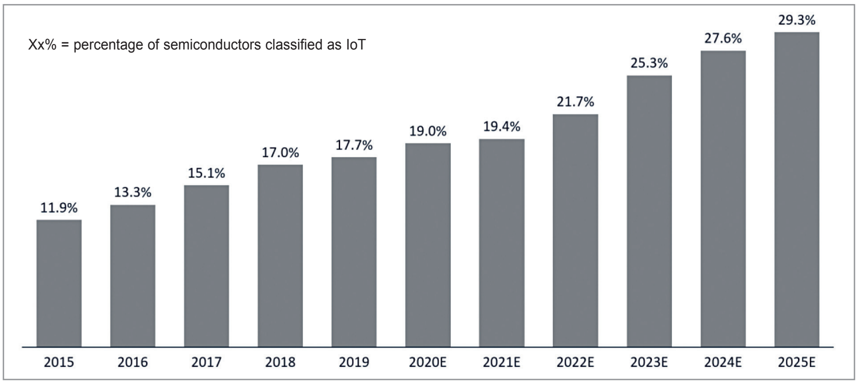

Fig. 2: Global IoT MCU penetration percentage

The current IoT device explosion will continue to drive the IoT semiconductor market and will likely lead to further semiconductor innovations amid higher production volumes. IoT Analytics latest report on this topic forecasts the IoT semiconductor component market to grow at a CAGR of 19% from US$33 billion in 2020 to US$80 billion in 2025. In particular, four components will be in the spotlight: IoT microcontrollers, IoT connectivity chipsets, IoT AI chipsets, and IoT security chipsets and modules.

Four key components driving IoT semiconductors

IoT Analytics defines IoT semiconductors as those that either individually or collaboratively contribute to the functionality of an IoT device or other IoT equipment. As such, several semiconductor components qualify as IoT semiconductors.

IoT Analytics research shows that the following four stand out in importance:

1. IoT microcontrollers (MCUs). In contrast to many other technology topics, IoT has a characteristically diverse set of use cases and applications. All these applications demand different levels of performance and functionality. Often, MCUs (rather than MPUs) are a good fit to deliver the required flexibility (for example, in terms of processing power) while at the same time presenting a very economical hardware choice for the application. An interesting trend is that IoT MCUs have now evolved from general-purpose MCUs and are sometimes becoming IoT application-specific MCUs that cater to the exact requirements posed by a specific IoT application (for example, in Q4 2020, NXP launched S32K3 automotive IoT MCU series).

On the back of these trends, IoT

research analysts and consulting team.

The market model for the 2015-25 period integrates publicly available information with intelligence IoT Analytics gathered from companies active in the market via telephonic interviews and personal meetings at business events during the last four years. When not known, regional and segment splits were approximated using several indicators such as general country KPIs (GDP, population, etc), number of companies supporting or adopting the technology in a specific region, or announcements, network rollouts, and deployments, etc.

Data from this report was compiled from a variety of primary and secondary data sources, including:

- Interviews with semiconductor companies' experts.
- Extensive secondary research of investor relation documents such as annual reports, press releases, white papers, company products and services portfolios, government and economic data, regulations, roadmaps, and industry case studies.
- We adopted the global semiconductor market data from World Semiconductor Trade Statistics for 2015 onwards up to November 2020.

Many industry gurus have been claiming that IoT is just a buzzword created by marketers and analysts. What would be your message to them?

In my opinion, IoT is not a buzzword anymore. IoT is in a state of evolution. Earlier, it was machine to machine (M2M) communication and going forward it will use artificial intelligence. Further, non-IoT devices contributed only to personal transformation, such as smartphones and tablets. IoT will bring social transformation like smart cities and smart healthcare.

If an investor were to sit with you and ask which are the top three or five types of tech that I should invest in—out of the entire gamut of tech involved in IoT—which would be your top favourites and why, specific to connectivity-related tech?

I would suggest the following technologies:

- Edge computing, which has given IoT a new perspective. Now it is not only about collecting data but computing data at the edge for real-time monitoring and insights. Further, the IoT ecosystem is already starting to shift from a cloud-centric approach to an edge.
- Artificial intelligence, which is already transforming smart IoT devices into intelligent IoT devices. Further, AI is already impacting many industries such as healthcare, automotive, and industrial IoT.
- IoT/cybersecurity, which is backbone of the IoT ecosystem. During Covid-19 Pandemic, whereas most businesses saw a sharp decline, the cybersecurity market saw huge growth.
- In terms of connectivity, cellular 5G IoT will have much potential to grow, especially after release 17 of 3GPP by 2022, covering high bandwidth application and low bandwidth.